

EMCH380 – Project Management Introduction

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Slides are developed from instructor's notes and other references.

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2. Role of the Project Manager
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Management Functions

Project Management – Introduction

OBJECTIVES OF THIS UNIT

- Define Projects
- Define Project Management
- Define the role and function of a project manager, line manager and the requirements for staffing projects under different organization structures
- Define the different management functions
- Highlight sources of conflicts in a project and knowledge of conflict handling modes.

DEFINITION OF A PROJECT

I

- A project can be considered to be any series of tasks/activities that:
 - Have a specific objective to be completed within certain specifications
 - Have defined start and end dates
 - Have funding limits (if applicable)
 - Consume resources (e.g., money, people equipment)

PROJECT MANAGEMENT

I

- Involves:
 - **Project Planning**
 - Definition of work requirements
 - Definition of quantity and quality of work
 - Definition of resources needed
 - Definition of when and how work requirements will be done
 - **Project monitoring and updating**
 - Tracking progress
 - Comparing actual outcome to planned outcome
 - Analyzing impact
 - Making adjustments

SUCCESSFUL PROJECT MANAGEMENT

I

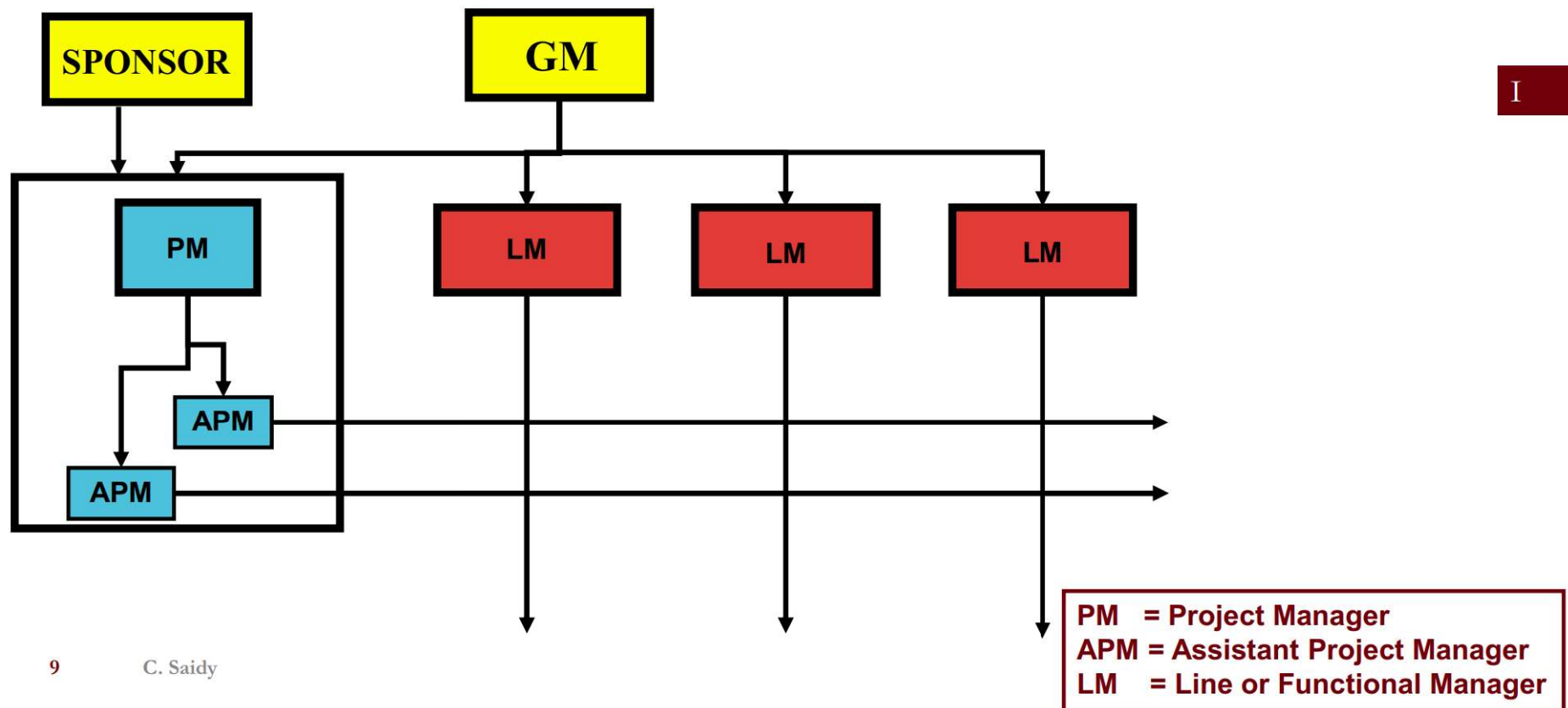
- Can be defined as having achieved the project objectives:
 - Within time
 - Within cost
 - At the desired performance/technology level
 - While utilizing the assigned resources effectively and efficiently
 - Accepted by the customer

SUCCESSFUL PROJECT MANAGEMENT (CONT'D)

I

**PROJECT MANAGEMENT IS THE ART OF
CREATING THE ILLUSION THAT ANY OUTCOME
IS THE RESULT OF A SERIES OF
PREDETERMINED, DELIBERATE ACTS WHEN IN
FACT IT WAS DUMB LUCK !**

MULTIPLE BOSS REPORTING



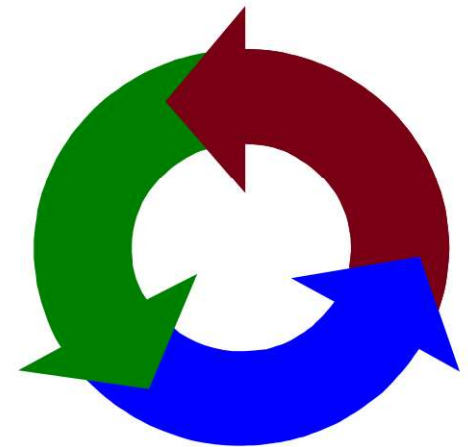


WHY USE PROJECT MANAGEMENT?

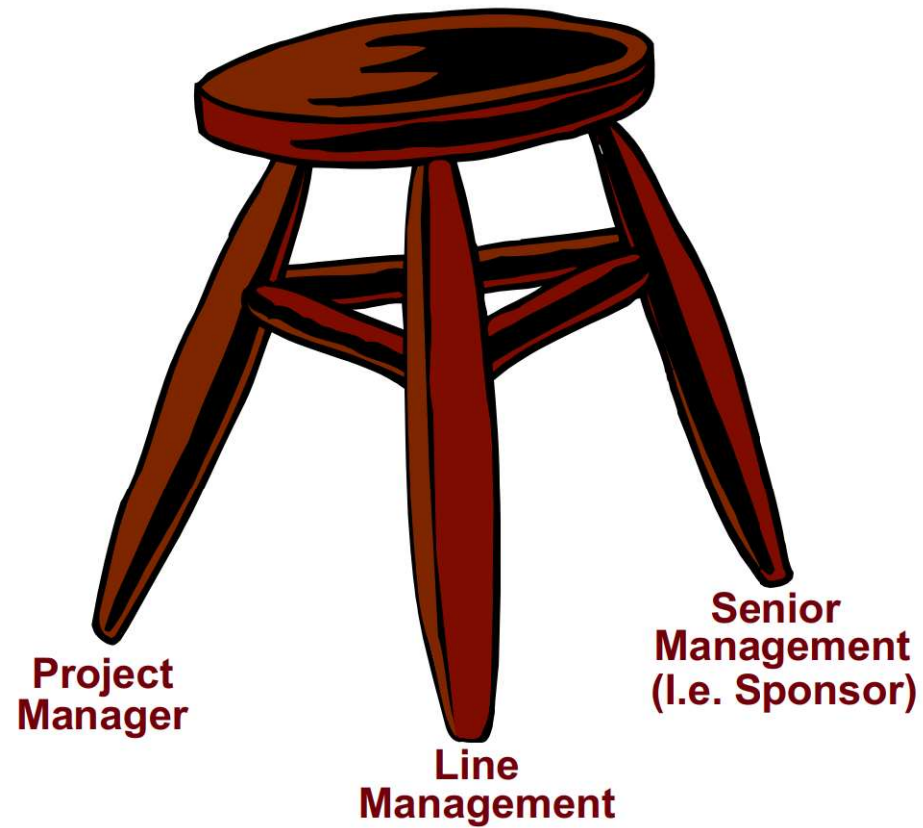
PROJECT MANAGEMENT MATURITY

I

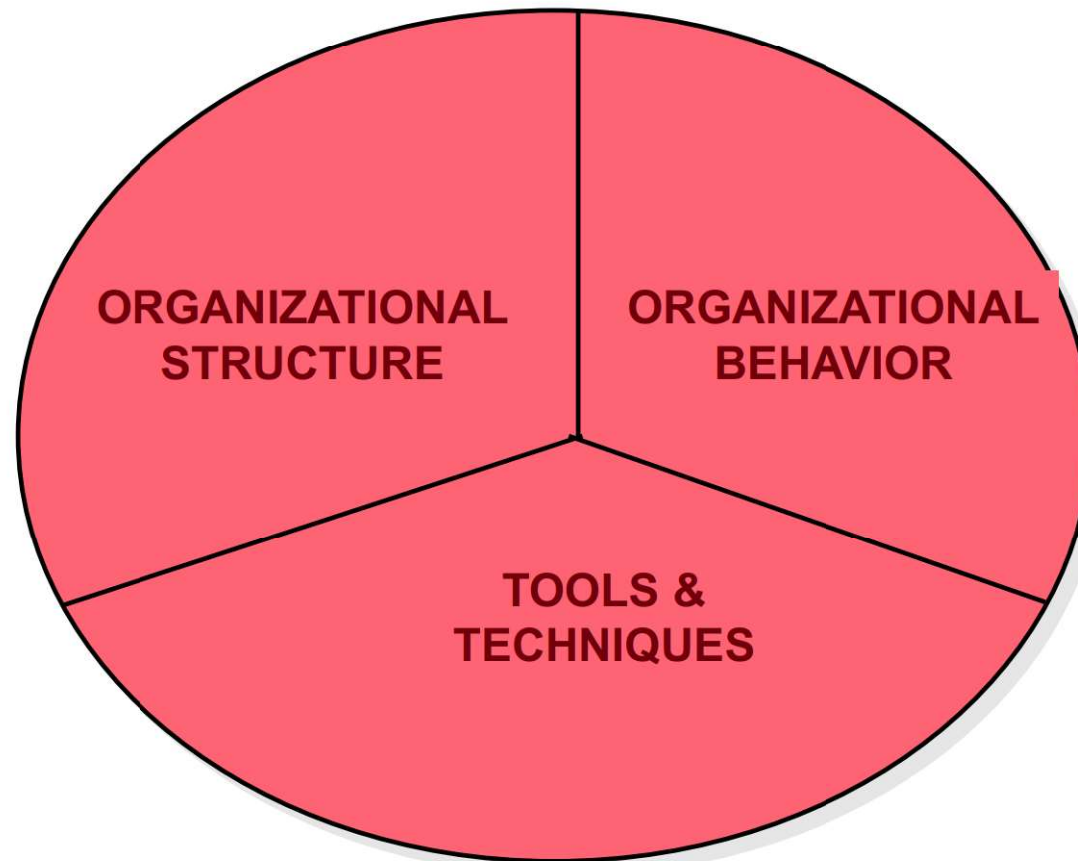
- Maturity in project management is like a three - legged stool.
- The legs represent the:
 - **Project manager**
 - **Line manager(s)**
 - **Executive management (i.e... Project sponsor)**
- Maturity cannot exist without stability



THE THREE-LEGGED STOOL



THE TOP OF THE THREE-LEGGED STOOL



EFFECTIVE PROJECT MANAGEMENT

I

- Effective project management requires an understanding of:
 - Quantitative tools and techniques
 - Organizational structures
 - Organizational behavior

Role of The Project Manager

Project Management – Introduction

ROLE OF THE PROJECT MANAGER

II

NEGOTIATING FOR RESOURCES

ROLE OF THE PROJECT MANAGER (CONT'D)

II

THE PROJECT KICKOFF MEETING

ROLE OF THE PROJECT MANAGER (CONT'D)

II

ESTABLISHING THE PROJECT'S POLICIES AND PROCEDURES

ROLE OF THE PROJECT MANAGER (CONT'D)

II

LAYING OUT THE PROJECT WORKFLOW AND PLAN

ROLE OF THE PROJECT MANAGER (CONT'D)

II

ESTABLISHING PERFORMANCE TARGETS

ROLE OF THE PROJECT MANAGER (CONT'D)

II

OBTAINING FUNDING

ROLE OF THE PROJECT MANAGER (CONT'D)

II

EXECUTING THE PLAN

ROLE OF THE PROJECT MANAGER (CONT'D)

II

ACTING AS THE CONDUCTOR

ROLE OF THE PROJECT MANAGER (CONT'D)

II

PUTTING OUT FIRES

ROLE OF THE PROJECT MANAGER (CONT'D)

II

COUNSELING & FACILITATION

ROLE OF THE PROJECT MANAGER (CONT'D)

II

ENCOURAGING THE TEAM TO FOCUS ON DEADLINES

ROLE OF THE PROJECT MANAGER (CONT'D)

II

MONITORING PROGRESS BY “POUNDING THE PAVEMENT”

ROLE OF THE PROJECT MANAGER (CONT'D)

II

EVALUATING PERFORMANCE

ROLE OF THE PROJECT MANAGER (CONT'D)

II

DEVELOP CONTINGENCY PLANS

ROLE OF THE PROJECT MANAGER (CONT'D)

II

BRIEFING THE PROJECT SPONSOR

ROLE OF THE PROJECT MANAGER (CONT'D)

II

BRIEFING THE TEAM

ROLE OF THE PROJECT MANAGER (CONT'D)

II

BRIEFING THE CUSTOMER

ROLE OF THE PROJECT MANAGER (CONT'D)

II

CLOSING OUT THE PROJECT

Classical Project Management

Project Management – Introduction

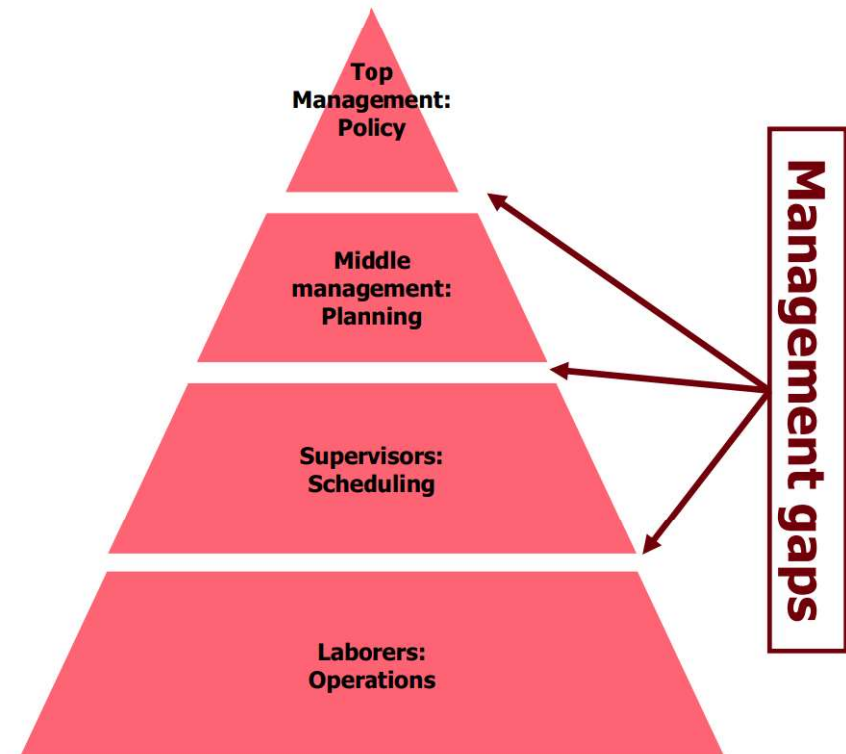
CLASSICAL MANAGEMENT

- Planning
- Organizing
- Staffing
- Controlling
- Directing

Which of the above is Usually NOT performed by the project manager?

STRUCTURE OF MANY COMPANIES

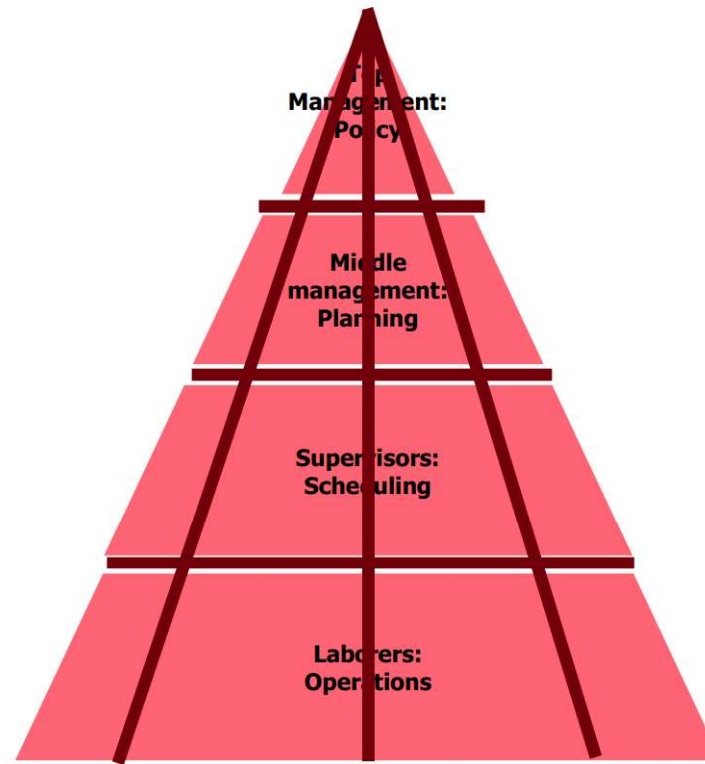
- There are always “class or prestige” gaps between various level of management (horizontal gaps)
- There are also functional gaps between working units of the organization (vertical gaps)



III

PROJECT MANAGERS FUNCTION

- Horizontal flow is the responsibility of the project managers, and their primary effort is to communicate and coordinate activities horizontally between the line organizations
- Project manager's role is to get these islands to communicate cross-functionally toward common goals and objectives.



III

DEFINITION OF PM REVISITED

III

- Project Management is the planning organizing directing and controlling of company resources for a relatively short-term objective that has been established to complete specific goals and objectives
- Project management uses the system's approach to management by having functional personnel (the vertical hierarchy) assigned to specific project (the horizontal hierarchy)

POTENTIAL BENEFITS FROM PM

- Identification of functional responsibilities to ensure that all activities are accounted for regardless of personnel turnover
- Identification of the time limits for scheduling
- Identification of a methodology for trade-off analysis
- Measurement of accomplishment against plans
- Early identification of problems so that corrective action may follow
- Improved estimating capability for future planning
- Knowing when objectives cannot be met or will be exceeded

PROJECT MANAGER – LINE MANAGER INTERFACE

III

- Project managers must control company resources within time, cost, and performance.
- Most companies have six resources:
 - Money
 - Manpower
 - Equipment
 - Facilities
 - Materials
 - Information/technology

PROJECT MANAGER – LINE MANAGER INTERFACE (CONT'D)

III

- Project Manager (PM) does not staff the project.
 - Staffing is a line responsibility
- The PM has the right to request specific resources, but the final decision of what resources will be committed rests with the line managers
- Long-term projects, which consume resources full-time, are usually set up as a separate division (if large enough) or as a line organization.

PROJECT MANAGER – LINE MANAGER INTERFACE (CONT'D)

III

- Actually the PM does not control any of these resources directly except perhaps for the money (or the budget).
- Resources are controlled by the line managers, functional managers, or as they are often called “resource managers”
- Project managers must negotiate with line managers for all project resources
- PM control these resources (which are temporarily loaned to them) through line managers.

PROJECT MANAGER – LINE MANAGER INTERFACE (CONT'D)

III

- Hence successful project management strongly depends on:
 - A good daily working relationship between project managers and line managers who directly assign resources to projects
 - Ability of functional employees to report vertically to their line manager at the same time that they report horizontally to one or more project managers

PROJECT MANAGER – LINE MANAGER INTERFACE (CONT'D)

III

- Potential problems:
 - Employees who report to multiple managers will always favor the manager who control their purse strings
 - Thus, most PMs appear to be at the mercy of line managers
 - When the PM gets in trouble, the only place he/she goes to is the line managers because additional resources are almost always required to alleviate the problem
 - When the line manager gets in trouble, he/she goes to the PM and requests either additional funding or some type of authorization for scope changes.

PROJECT MANAGER'S ROLE

III

- Project Manager is responsible for coordinating and integrating activities across multiple functional lines
- To do this:
 - PM needs strong communication and interpersonal skills
 - Must become familiar with the operations of each line organization
 - Should have good knowledge of the technology being used.

PROJECT MANAGER'S ROLE (CONT'D)

III

- Project manager's may have increasing responsibility but very little authority
- Lack of authority can force them to negotiate with upper-level management as well as functional management for control of company resources

PROJECT MANAGER'S ROLE (CONT'D)

III

- An executive with a computer manufacturer who was looking to hire a project manager stated that:

“I can teach people the technology and give them technical experts to assist them in decision making but I cannot teach somebody how to work with people”

INTERFACE MANAGEMENT

III

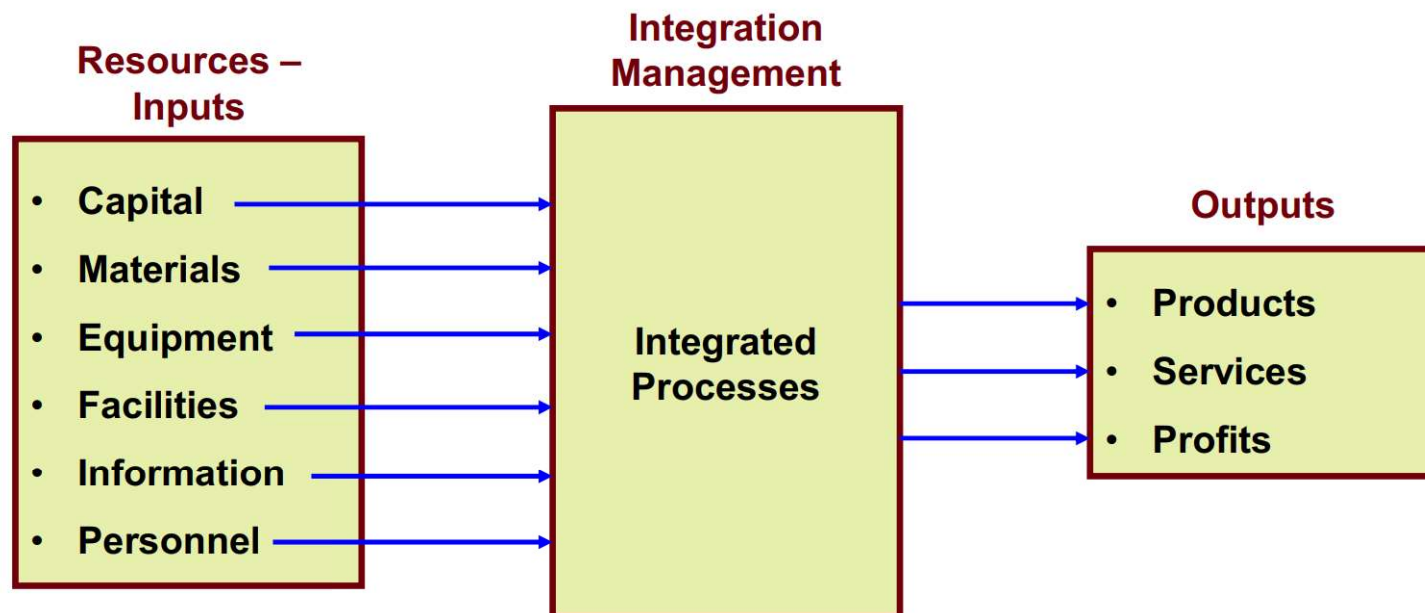
- Managing human interrelationships within the project team
- Managing human interrelationships between the project team and the functional organization
- Managing human interrelationships between the project team and senior management
- Managing human interrelationships between the project team and the customer's organization, whether an internal or external organization

INTERFACE MANAGEMENT (CONT'D)

III

As part of interface management, the project manager's role also includes integration management.

INTEGRATION MANAGEMENT



III

THE FUNCTIONAL ROLE

III

- The functional manager has the responsibility to define how the task will be done and where the task will be done (i.e., the technical criteria)
- The functional manager has the responsibility to provide sufficient resources to accomplish the objective within the project's constraints (i.e., who will get the job done).

FUNCTIONAL OBSTACLES

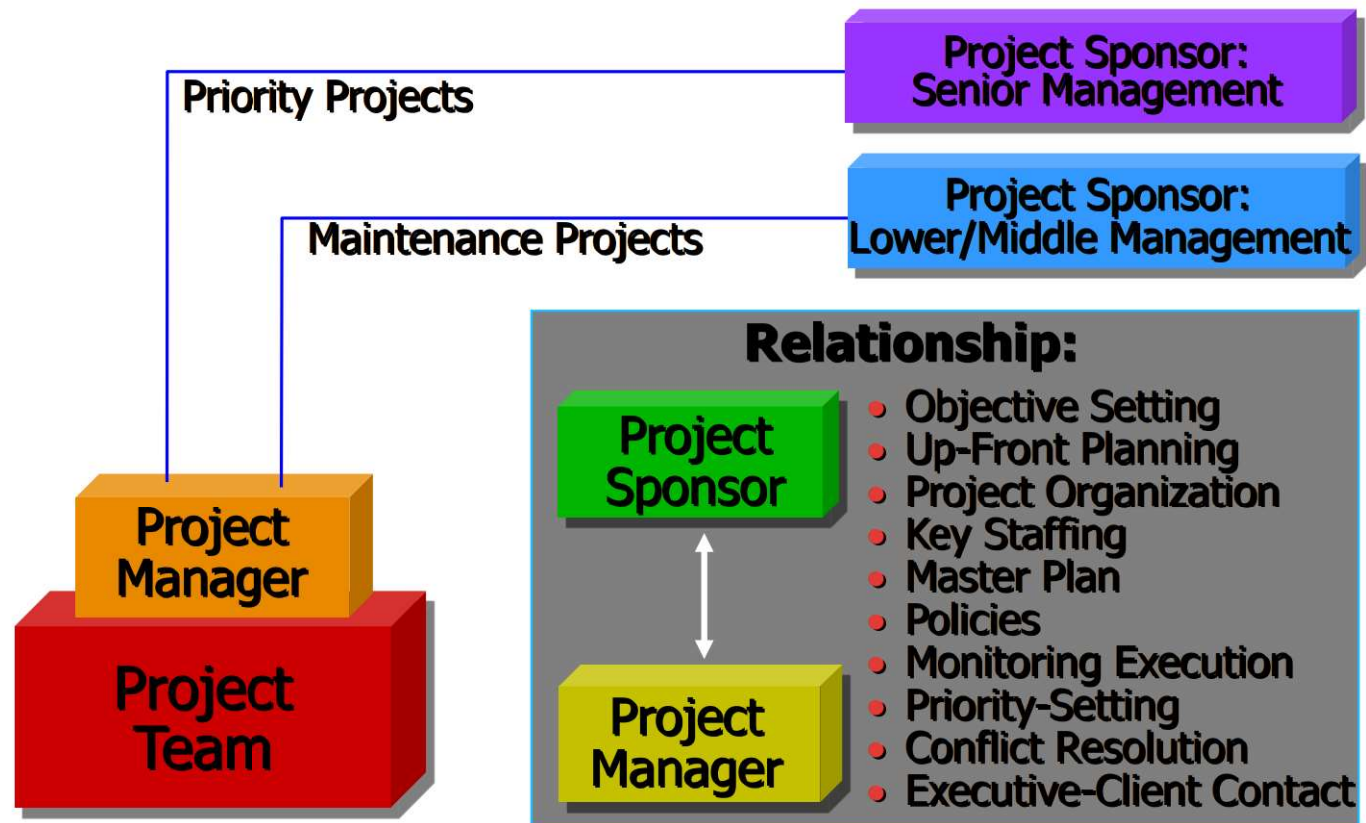
- Unlimited work requests (especially during competitive bidding)
- Predetermined deadlines
- All requests having a high priority
- Limited number of resources
- Limited availability of resources
- Unscheduled changes in the project plan
- Unpredicted lack of progress

FUNCTIONAL OBSTACLES

III

Most projects also have a project sponsor which may or may not reside at the executive levels of management.

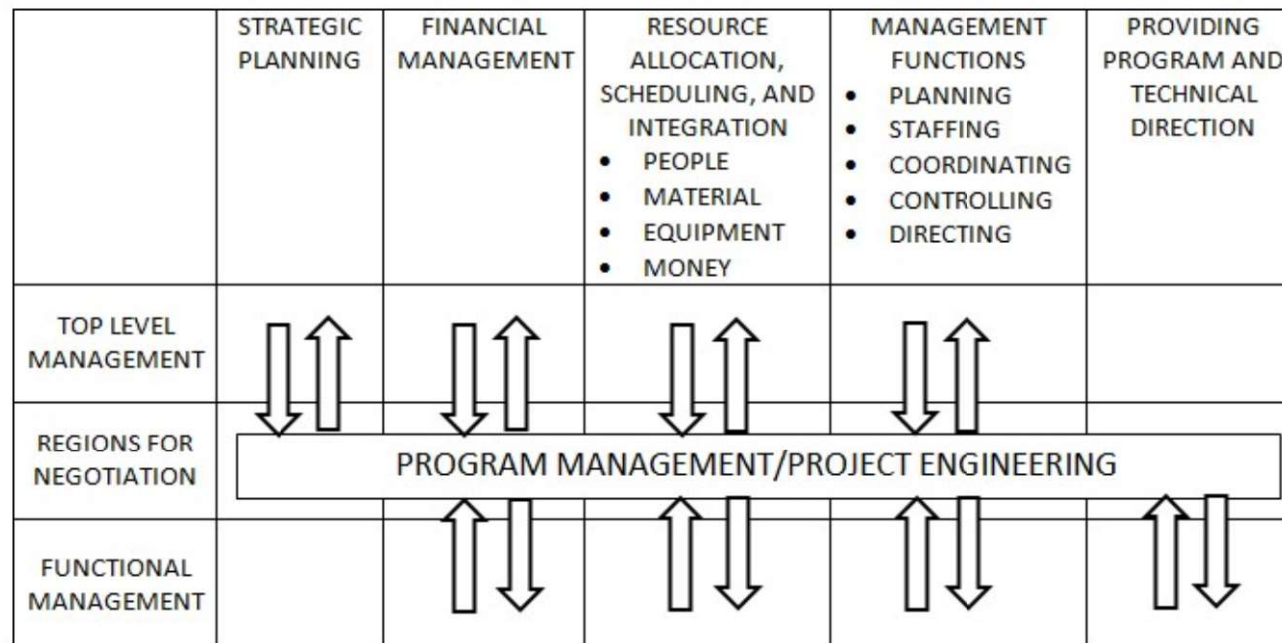
THE PROJECT SPONSOR INTERFACE



III

NEGOTIATING OF SYSTEMS ACTIVITIES

III



The negotiation activities of systems management

PM – LM INTERFACE SUMMARY

III

- Project manager must maintain the balance between technical and managerial project function and survive organizational constraints
- Organizational constraints have a tendency to develop into organizational conflict
- This requires that top management take an active role in conflict resolution by establishing priorities among projects.

PM – LM INTERFACE SUMMARY (CONT'D)

III

- Project management is designed to have shared authority and responsibility between the project and line managers
- Project managers plan monitor and control the project
- Whereas functional managers perform the work

PROJECT VS NON-PROJECT-DRIVEN ORGANIZATIONS

III

- At the micro level all organizations are either marketing-, engineering-, or manufacturing-driven
- At the macro level organizations are either project or non project-driven organizations

PROJECT VS NON-PROJECT-DRIVEN ORGANIZATIONS (CONT'D)

III

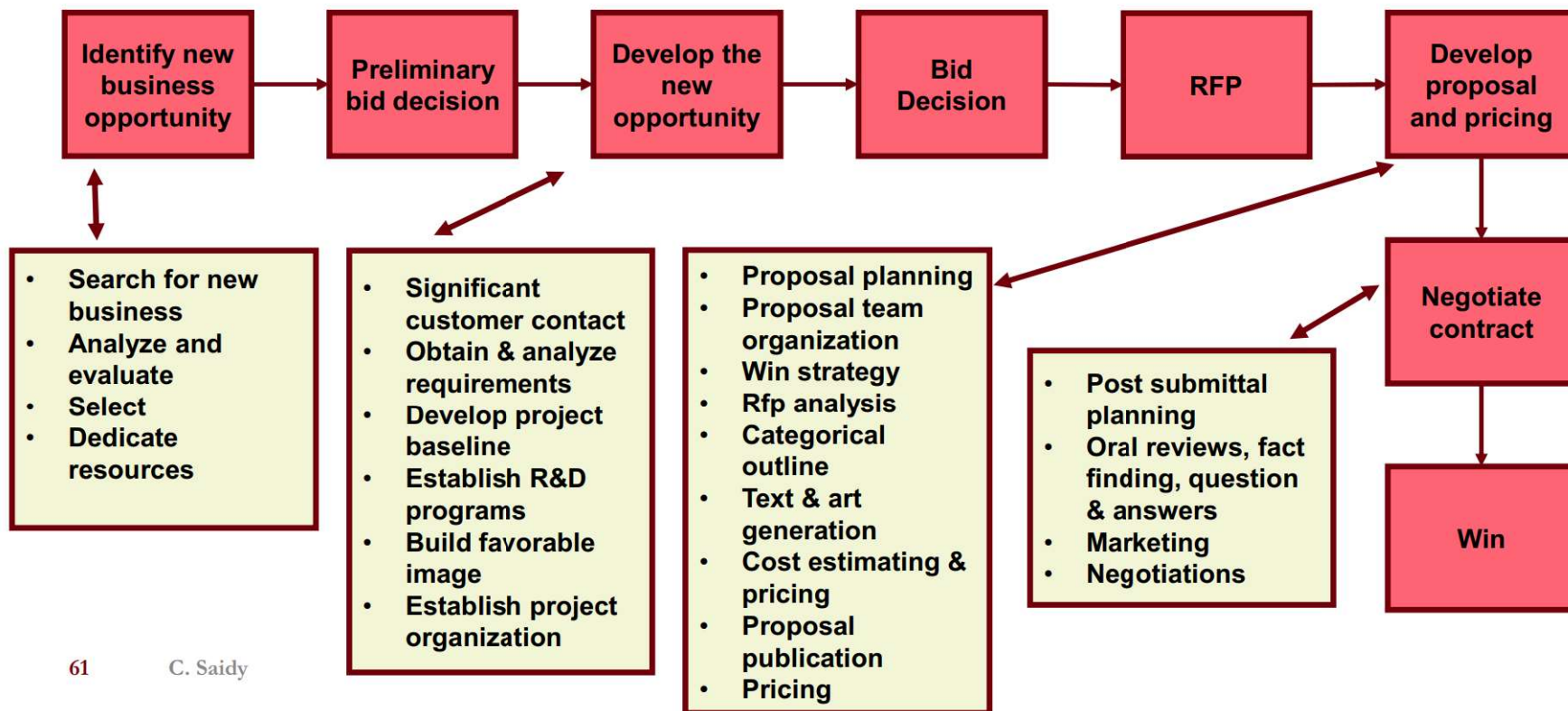
- In non-project-driven organization such as low-technology manufacturing, profit and loss are measured on vertical or functional lines
- In these organizations, projects exist to support the product lines or the functional lines
- Project management in a non-project driven organization is generally more difficult than in project-driven

PROJECT VS NON-PROJECT-DRIVEN ORGANIZATIONS (CONT'D)

III

- At the micro level all organizations are either marketing-, engineering-, or manufacturing-driven
- At the macro level organizations are either project or non project-driven organizations

PHASES OF WINNING NEW CONTRACTS IN PROJECT-DRIVEN ORGANIZATIONS)



III

Thank you for your attention!

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